19			38.6	5	P1 process to find CD, e.g. $\tan 34 = \frac{CD}{20}$ P1 process to find BD, e.g. $\cos 34 = \frac{20}{BD}$ or $BD^2 = 20^2 + \text{“}CD\text{”}^2$ P1 uses sine rule to find AD e.g. $\frac{AD}{\sin 60} = \frac{\text{ital } BD}{\sin(180-60-45)}$ P1 process to find DAC, e.g. $\sin DAC = \frac{\text{ital } DC}{\text{ital } AD}$ A1 answer in range 38.5 – 38.6

Documented scheme clarification

1MA1_MOCK_SET_01_3H_Q19

Original source

scheme page 12, SHA256 02e301a240422cbf615b085af66b5688c19aa7f3ae5a08b6dcf28f9daaf67ec7

Original wording

The final trigonometric method line prints tan DAC = DC/AD.

Correction / interpretation

Use sin DAC = DC/AD because AD is the hypotenuse of right-angled triangle ACD. With CD = 20 tan 34 degrees, BD = 20/cos 34 degrees and AD = BD sin 60 degrees/sin 75 degrees, angle DAC is 38.6 degrees to one decimal place. The source final answer is correct.

Reason

CD is opposite angle DAC and AD is the hypotenuse. The printed tangent ratio is therefore a source method-line error.

Sol authority

255ab3b5e61075efcd15bf38f84a9b1c3db32548995214bca495d2a2935b5569